



# basic education

Department:  
Basic Education  
**REPUBLIC OF SOUTH AFRICA**

**SENIOR CERTIFICATE EXAMINATIONS/  
NATIONAL SENIOR CERTIFICATE EXAMINATIONS  
*SENIORSERTIFIKAAT-EKSAMEN/  
NASIONALE SENIORSERTIFIKAAT-EKSAMEN***

**MATHEMATICAL LITERACY P2/*WISKUNDIGE GELETTERDHEID V2***

**MAY/JUNE/*MEI/JUNIE* 2026**

**MARKING GUIDELINES/*NASIENRIGLYNE***

**MARKS/*PUNTE*: 150**

| <b>Symbol/Kode</b> | <b>Explanation/<i>Verduideliking</i></b>                                                                                                                                  |
|--------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>MA</b>          | Method with accuracy/ <i>Metode met akkuraatheid</i>                                                                                                                      |
| <b>MCA</b>         | Method with consistent accuracy/ <i>Metode met volgehoue akkuraatheid</i>                                                                                                 |
| <b>CA</b>          | Consistent accuracy/ <i>Volgehoue akkuraatheid</i>                                                                                                                        |
| <b>A</b>           | Accuracy/ <i>Akkuraatheid</i>                                                                                                                                             |
| <b>C</b>           | Conversion/ <i>Herleiding</i>                                                                                                                                             |
| <b>S</b>           | Simplification/ <i>Vereenvoudiging</i>                                                                                                                                    |
| <b>RT</b>          | Reading from a table/graph/document/diagram/ <i>Lees vanaf tabel/grafiek/dokument/diagram</i>                                                                             |
| <b>SF</b>          | Correct substitution in a formula/ <i>Korrekte vervanging in 'n formule</i>                                                                                               |
| <b>O</b>           | Opinion/Explanation/ <i>Opinie/Verduideliking</i>                                                                                                                         |
| <b>P</b>           | Penalty, e.g. for no units, incorrect rounding off, etc./ <i>Penalising, bv. vir geen eenhede, verkeerde afronding, ens.</i>                                              |
| <b>NPR</b>         | No penalty for correct rounding/ <i>Geen penalising vir korrekte afronding nie</i>                                                                                        |
| <b>NPU</b>         | No penalty for omitting unit, but wrong unit is penalised/ <i>Geen penalisinge indien die eenheid uitgelos is nie, maar wel indien 'n verkeerde eenheid gebruik word.</i> |
| <b>AO</b>          | Answer only/ <i>Slegs antwoord</i>                                                                                                                                        |

**These marking guidelines consist of 16 pages./  
*Hierdie nasienriglyne bestaan uit 16 bladsye.***

**NOTE:**

- If a candidate answers a question TWICE, only mark the FIRST attempt.
- If a candidate has crossed out (cancelled) an attempt to a question and NOT redone the solution, mark the crossed out (cancelled) version.
- Consistent accuracy (CA) applies in ALL aspects of the marking guidelines; however it stops at the second calculation error.
- If the candidate presents any extra solution when reading from a graph, table, layout plan and map, then penalise for every extra item presented.
- Rounding is an independent mark.
- General principle of marking, if the candidate makes one mistake one mark is deducted.
- A conclusion mark can only be given if  $\frac{1}{3}$  of the total marks for the sub-question have been awarded.
- No penalty for rounding (NPR) if the first decimal is correct.

**LET WEL:**

- As 'n kandidaat 'n vraag TWEE KEER beantwoord, sien slegs die EERSTE poging na.
- As 'n kandidaat 'n antwoord van 'n vraag doodtrek (kanselleer) en nie oordoen nie, sien die doodgetrekte (gekanselleerde) poging na.
- Volgehoue akkuraatheid (CA) word in ALLE aspekte van die nasienriglyne toegepas; dit hou egter op by die tweede berekeningsfout.
- Wanneer 'n kandidaat aflesings vanaf 'n grafiek, tabel, uitlegplan en kaart aanbied en ekstra antwoorde gee, penaliseer vir elke ekstra item.
- Afronding tel as 'n afsonderlike punt.
- Die algemene beginsel van nasien is, as 'n leerder een fout maak, word een punt afgetrek.
- 'n Gevolgtrekkingspunt kan slegs gegee word indien  $\frac{1}{3}$  van die totale punte vir die subvraag toegeken is.
- Geen penalisering vir afronding (NPR) nie as die eerste desimaal korrek is.

| <b>QUESTION/VRAAG 1 [30 MARKS/30 PUNTE] ANSWER ONLY FULL MARKS</b> |                                                          |                                   |                |
|--------------------------------------------------------------------|----------------------------------------------------------|-----------------------------------|----------------|
| <b>Q/V</b>                                                         | <b>Solution/Oplossing</b>                                | <b>Explanation/Verduideliking</b> | <b>T&amp;L</b> |
| 1.1.1                                                              | ✓✓ A<br>Metric system/Metrieke stelsel                   | 2A correct concept<br>(2)         | M<br>L1<br>E   |
| 1.1.2                                                              | ✓✓ A<br>Strip chart/Strookkaart                          | 2A correct concept<br>(2)         | MP<br>L1<br>E  |
| 1.1.3                                                              | ✓✓ A<br>Layout plan/Uitlegplan                           | 2A correct concept<br>(2)         | MP<br>L1<br>E  |
| 1.1.4                                                              | ✓✓ A<br>Probability/Waarskynlikheid                      | 2A correct concept<br>(2)         | P<br>L1<br>E   |
| 1.1.5                                                              | ✓✓ A<br>Diameter / Middellyn                             | 2A correct concept<br>(2)         | M<br>L1<br>E   |
| 1.2.1                                                              | 7 ✓✓ RT                                                  | 2RT reading from table<br>(2)     | M<br>L1<br>E   |
| 1.2.2                                                              | 50% ✓✓ RT<br>Saturday/Saterdag <b>OR/OF</b> SAT./SA. ✓RT | 2RT percentage<br>1RT day<br>(3)  | P<br>L1<br>E   |

| Q/V   | Solution/Oplissing                                                                                                                                                                                         | Explanation/Verduideliking                                                               | T&L           |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------------------------------------------------------------------------|---------------|
| 1.2.3 | ✓✓RT<br>Wednesday/ <i>Woensdag</i> <b>OR/OF</b> WED. /WO                                                                                                                                                   | 2RT reading from table<br>(2)                                                            | M<br>L1<br>E  |
| 1.2.4 | ✓MA<br>1,3 mm + 1,4 mm + 18 mm<br><br>= 20,7 mm ✓A                                                                                                                                                         | 1MA addition of values<br><br>1A simplification<br>(2)                                   | M<br>L1<br>E  |
| 1.2.5 | ✓A ✓A<br>Seven minutes to three in the afternoon<br><i>Sewe minute voor drie in die middag</i><br><br>✓A <b>OR/OF</b><br>Fifty-three minutes past two p.m. ✓A<br><i>Drie en vyftig minute na twee n.m.</i> | 1A correct words<br>1A afternoon<br>(2)                                                  | M<br>L1<br>E  |
| 1.3.1 | ✓✓A<br>Elevation map<br><i>Hoogtekaart</i>                                                                                                                                                                 | 2A correct concept<br>(2)                                                                | MP<br>L1<br>E |
| 1.3.2 | 1 750 m ✓✓RT                                                                                                                                                                                               | 2RT reading from map<br>[Accept answers more than<br>1700 m but less than 1780 m]<br>(2) | MP<br>L1<br>E |
| 1.3.3 | D ✓✓RT                                                                                                                                                                                                     | 2RT reading from map<br>(2)                                                              | MP<br>L1<br>E |
| 1.3.4 | ✓RT ✓RT<br>16 km – 5 km<br>= 11 km ✓A                                                                                                                                                                      | 1RT 16 km<br>1RT 5 km<br>1A simplification<br><br>[Accept 10,8 – 11,2]<br>(3)            | MP<br>L1<br>M |
|       |                                                                                                                                                                                                            | [30]                                                                                     |               |

| <b>QUESTION 2 [28 MARKS/28 PUNTE]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |                                                                                                                                                                                                                                                                                                                                                                                                                                        |                |
|---------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Q/V</b>                            | <b>Solution/Oplissing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | <b>Explanation/Verduideliking</b>                                                                                                                                                                                                                                                                                                                                                                                                      | <b>T&amp;L</b> |
| 2.1.1                                 | ✓✓ A<br>Bar scale/Staafskaal                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2A bar scale<br>(2)                                                                                                                                                                                                                                                                                                                                                                                                                    | MP<br>L1<br>E  |
| 2.1.2                                 | R37 ✓✓ RT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 2RT correct number<br>(2)                                                                                                                                                                                                                                                                                                                                                                                                              | MP<br>L1<br>E  |
| 2.1.3                                 | Hazyview ✓✓ A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 2A correct destination<br>(2)                                                                                                                                                                                                                                                                                                                                                                                                          | MP<br>L2<br>M  |
| 2.1.4                                 | 14 ✓✓ RT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2RT correct number<br>(2)                                                                                                                                                                                                                                                                                                                                                                                                              | MP<br>L1<br>M  |
| 2.1.5                                 | <p>✓A<br/>Scale/Skaal 33 mm = 6 km</p> <p>✓A<br/>Map distance/Kaartafstand 83 mm</p> <p>Real distance/Werklike afstand<br/> <math>= \frac{83}{33} \times 6 \text{ km}</math> ✓MA<br/> <math>= 15,09... \text{ km}</math> ✓CA<br/> <math>\approx 15 \text{ km}</math> ✓R</p> <p>✓A <b>OR/OF</b><br/> Scale/Skaal 11 mm = 2 km</p> <p>Map distance/Kaartafstand 83 mm ✓A</p> <p>Real distance/Werklike afstand<br/> <math>= \frac{83}{11} \times 2 \text{ km}</math> ✓MA<br/> <math>= 15,09... \text{ km}</math> ✓CA<br/> <math>\approx 15 \text{ km}</math> ✓R</p> <p>✓A <b>OR/OF</b><br/> 3,3 cm = 6 km<br/> 3,3 cm = 600 000 cm<br/> 1 : 181 818,18181</p> <p>✓A ✓MA<br/> <math>8,1 \text{ cm} \times 181,818181 = 1\,472\,727,272 \text{ cm}</math><br/> <math>= 14,7 \text{ km}</math> ✓CA<br/> <math>\approx 15 \text{ km}</math> ✓R</p> | <p>1A measured scale</p> <p>1A map distance</p> <p>1MA using the ratio</p> <p>1CA simplification<br/>1R correct rounding</p> <p><b>OR/OF</b></p> <p>1A measured scale</p> <p>1A map distance</p> <p>1MA using the ratio</p> <p>1CA simplification<br/>1R correct rounding</p> <p><b>OR/OF</b></p> <p>1A measured scale</p> <p>1A map distance<br/>1MA using the ratio</p> <p>1CA simplification<br/>1R correct rounding</p> <p>(5)</p> | MP<br>L3<br>M  |

| Q/V   | Solution/Oplissing                                                                                                                                                                                                                                                | Explanation/Verduideliking                                                                                                                                                              | T&L           |
|-------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 2.2.1 | 11 ✓✓ A                                                                                                                                                                                                                                                           | 2A correct number<br>(2)                                                                                                                                                                | MP<br>L1<br>E |
| 2.2.2 | Number on lawn/Getal op grasperk<br>✓ MA ✓ RT<br>= 16 000 – (5 500 + 3 500)<br>= 7 000 ✓ CA                                                                                                                                                                       | 1MA subtracting<br>1RT correct values (at least 2)<br><br>1CA simplification<br>(3)                                                                                                     | MP<br>L2<br>E |
| 2.2.3 | NW <b>OR/OF</b> North-west/Noordwes ✓✓ A                                                                                                                                                                                                                          | 2A correct direction<br>(2)                                                                                                                                                             | MP<br>L2<br>M |
| 2.2.4 | Distance/Afstand<br><br>✓ RT<br>= $\frac{100}{3} \times 91,44$ cm ✓ MA<br>= 3 048 cm ✓ CA<br>= 30,48 m ✓ C<br><br><b>OR/OF</b><br><br>3 feet = 91,44 cm<br>1 foot = 30,48 cm<br>✓ RT<br>100 feet = $100 \times 30,48$ cm ✓ MA<br>= 3 048 cm ✓ CA<br>= 30,48 m ✓ C | 1RT 100 ft<br>1MA using the ratio<br><br>1CA simplification<br>1C conversion<br><br><b>OR/OF</b><br><br>1RT 100 ft<br>1MA using the ratio<br>1CA simplification<br>1C conversion<br>(4) | MP<br>L2<br>D |
| 2.2.5 | Last row in ZZ/Laaste ry in ZZ<br>✓ MA<br>= 20 + 23 ✓ RT<br>= 43 ✓ CA<br><br>INVALID / NIE GELDIG NIE ✓ O                                                                                                                                                         | 1MA addition<br>1RT 23<br>1CA simplification<br><br>1O verification<br>(4)                                                                                                              | MP<br>L4<br>M |
|       |                                                                                                                                                                                                                                                                   | [28]                                                                                                                                                                                    |               |

| <b>QUESTION 3 [31 MARKS]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |                                                                                                                                                                                                                                                             |                |
|------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Q/V</b>                   | <b>Solution/Oplissing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <b>Explanation/Verduideliking</b>                                                                                                                                                                                                                           | <b>T&amp;L</b> |
| 3.1.1                        | $D = (90 \text{ m} - 40,3 \text{ m}) \div 2$ $= 24,85 \text{ m} \quad \checkmark \text{CA}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 1MA subtract and dividing by 2<br>1CA simplification<br><b>AO</b><br>(2)                                                                                                                                                                                    | M<br>L2<br>E   |
| 3.1.2                        | <p>Area of one penalty box/<i>Opp. van een doelgebied</i></p> $= 40,3 \text{ m} \times 16,5 \text{ m} \quad \checkmark \text{SF}$ $= 664,95 \text{ m}^2 \quad \checkmark \text{A}$ <p>Area of soccer field/<i>Oppervlakte van sokkerveld</i></p> $= 120 \text{ m} \times 90 \text{ m}$ $= 10\,800 \text{ m}^2 \quad \checkmark \text{A}$ <p>Percentage/<i>Persentasie</i> = <math>\frac{664,95}{10\,800} \times 100\%</math> <math>\checkmark \text{MCA}</math></p> $= 6,156944 \% \quad \checkmark \text{CA}$ <p><b>OR/OF</b></p> <p>Area of penalty box ÷ area of soccer field/<br/><i>Opp. van doelgebied ÷ opp. van sokkerveld</i></p> $= \frac{40,3 \text{ m}}{120 \text{ m}} \times \frac{16,5 \text{ m}}{90 \text{ m}} \quad \checkmark \text{SF} \quad \checkmark \text{A} \quad \checkmark \text{A}$ <p>Percentage/<i>Persentasie</i> = <math>0,335833... \times 0,18333 \times 100\%</math></p> $= 6,156944\% \quad \checkmark \text{CA}$ | 1SF substitution<br>1A simplification<br>1A simplification<br>1MCA correct fraction<br>1CA simplification<br><b>OR/OF</b><br>1MCA correct fraction<br>1SF substitution<br>1A simplification<br>1A simplification<br>1CA simplification<br><b>NPR</b><br>(5) | M<br>L3<br>M   |
| 3.1.3                        | $\text{Perimeter/Omtrek} = 2 \times 120 \text{ m} + 2 \times 90 \text{ m}$ $= 420 \text{ m} \quad \checkmark \text{A}$ $\text{Distance ran/Afstand afgelê} = 10 \times 420 \text{ m} = 4\,200 \text{ m}$ $= 4,2 \text{ km} \quad \checkmark \text{C}$ <p>INVALID/<i>NIE GELDIG NIE.</i> <math>\checkmark \text{O}</math></p> <p><b>OR/OF</b></p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | 1SF substitution in formula<br>1A simplification<br>1CA multiplying correct values<br>1C conversion<br>1O conclusion<br><b>OR/OF</b>                                                                                                                        | M<br>L4<br>M   |

| Q/V   | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                | T&L          |
|-------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
|       | <p>Length distance for 10 laps/<i>Lengte-afstand vir 10 rondtes</i><br/> <math>= 120 \text{ m} \times 2 \times 10</math><br/> <math>= 2\,400 \text{ m}</math> ✓ A</p> <p>Width distance for 10 laps/<i>Breedte-afstand vir 10 rondtes</i><br/> <math>= 90 \text{ m} \times 2 \times 10</math><br/> <math>= 1\,800 \text{ m}</math> ✓ A</p> <p>Total distance/<i>Totale afstand</i> <math>= 2\,400 \text{ m} + 1\,800 \text{ m}</math><br/> <math>= 4\,200 \text{ m}</math> ✓ CA</p> <p>Distance in km/<i>Afstand in km</i> <math>= 4\,200 \div 1\,000 = 4,2</math> km ✓ C</p> <p>INVALID/NIE GELDIG NIE ✓ O</p> <p style="text-align: center;"><b>OR/OF</b><br/>✓ SF</p> <p>Perimeter/<i>Omtrek</i> <math>= 2 \times 120 \text{ m} + 2 \times 90 \text{ m}</math><br/> <math>= 420 \text{ m}</math> ✓ A</p> <p>Conversion to m/<i>Herleiding na m:</i><br/> <math>4 \text{ km} = 4\,000 \text{ m}</math> ✓ C</p> <p>Distance for 1 lap/<i>Afstand vir 1 rondte</i><br/> <math>= 4\,000 \text{ m} \div 10</math><br/> <math>= 400 \text{ m}</math> ✓ CA</p> <p>INVALID/NIE GELDIG NIE ✓ O</p> | <p>1A correct values for length</p> <p>1A correct values for width</p> <p>1CA simplification</p> <p>1C conversion</p> <p>1O conclusion</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1SF substitution in formula</p> <p>1A simplification</p> <p>1C conversion</p> <p>1CA dividing correct values</p> <p>1O conclusion</p> <p style="text-align: right;">(5)</p> |              |
| 3.2.1 | <p><math>P = \frac{1}{18}</math> ✓ A</p> <p><math>\approx 0,056</math> ✓ R</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | <p>1A numerator</p> <p>1A denominator</p> <p>1R correctly rounded</p> <p style="text-align: right;">(3)</p>                                                                                                                                                                                                                                                               | P<br>L2<br>E |
| 3.2.2 | <p>Diameter/<i>Middel lyn</i> <math>= 1,2 \text{ m} \times 2 = 2,4 \text{ m}</math> ✓ MA</p> <p>Total distance/<i>Totale afstand</i> <math>= 3,142 \times 2,4 \text{ m}</math> ✓ SF</p> <p><math>= 7,5408 \text{ m}</math> ✓ CA</p> <p style="text-align: center;"><b>OR/OF</b><br/>✓ MA</p> <p>Total distance/<i>Totale afstand</i> <math>= 3,142 \times 1,2 \text{ m} \times 2</math> ✓ SF</p> <p><math>= 7,5408 \text{ m}</math> ✓ CA</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | <p>1MA calculate diameter</p> <p>1SF correct substitution</p> <p>1CA simplification</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1MA calculate diameter</p> <p>1SF correct substitution</p> <p>1CA simplification</p> <p style="text-align: right;">(3)</p>                                                                                                     | M<br>L2<br>M |
| 3.2.3 | Direct/ <i>Direkte</i> ✓✓ A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  | 2A direct                                                                                                                                                                                                                                                                                                                                                                 | M<br>L1<br>E |

| Q/V          | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | T&L          |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------|
| 3.2.4<br>(a) | <p>Radius = 5 m ÷ 2<br/>= 2,5 m ✓ MA</p> <p>Volume = 3,142 × (2,5 m)<sup>2</sup> × 0,21 m<br/>= 4,123875 m<sup>3</sup> ✓ SF ✓ C</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | <p>1MA radius</p> <p>1SF substitution of radius<br/>1C conversion to metres</p> <p>(3)</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | M<br>L2<br>M |
| 3.2.4<br>(b) | <p>Litres per minute = 5 × 3,78541178 ✓ MA<br/>= 18,9270589 ℓ ✓ A</p> <p>Volume = 4,123875 m<sup>3</sup> × 1 000 = 4 123,875 ℓ ✓ C</p> <p>Number of minutes/Aantal minute = <math>\frac{4\,123,875\,\ell}{18,9270589\,\ell/m}</math> ✓ MCA<br/>= 217,8825...min ✓ CA</p> <p>Time/Tyd = 217,8825... ÷ 60 ✓ MCA<br/>= 3,63... ✓ CA<br/>= 3 hours/uur 38 minutes/minute ✓ R</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Volume = 4,123875 m<sup>3</sup> × 1 000 = 4 123,875 ℓ ✓ C</p> <p>Number of gallons/Aantal gellings = <math>\frac{4\,123,875}{3,78541178}</math> ✓ MCA<br/>= 1 089,413844 ✓ CA</p> <p>Number of minutes/Aantal minute = 1 089,413844 ÷ 5 ✓ MA<br/>= 217,882... ✓ CA ✓ MCA</p> <p>Time in hours and min/Tyd in ure en min = 217,882 ÷ 60<br/>= 3,63... ✓ CA<br/>= 3 hours/uur 38 minutes/minute ✓ R</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Number of litres/min/Aantal liters/min<br/>✓ MA<br/>= 5 × 3,7841178 ℓ = 18,9270589 ℓ ✓ A</p> <p>Volume = 18,9270589 ℓ ÷ 1000 = 0,0189271 m<sup>3</sup> ✓ C</p> <p>Number of minutes/Aantal minute = <math>\frac{4\,123,875}{0,0189271}</math> ✓ MCA<br/>= 217,8825047 ✓ CA</p> <p>Time/Tyd = 217,8825047 ÷ 60 ✓ MCA<br/>= 3 hours/uur 38 minutes/minute ✓ CA ✓ R</p> | <p>1MA multiplication<br/>1A simplification litres<br/>1C conversion</p> <p>1MCA division<br/>1CA simplification<br/>1MCA divide minutes by 60<br/>1CA simplification<br/>1R rounding of minutes</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1C conversion<br/>1MCA dividing<br/>1CA simplification<br/>1MA division by 5<br/>1CA simplification<br/>1MCA divide minutes by 60</p> <p>1CA simplification<br/>1R rounding of minutes</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1MA multiplication<br/>1A simplification litres<br/>1C conversion<br/>1MCA division<br/>1CA simplification<br/>1MCA divide minutes by 60<br/>1CA simplification<br/>1R rounding of minutes</p> <p>(8)</p> | M<br>L3<br>D |
|              |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | [31]         |

| <b>QUESTION 4 [33 MARKS]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |                                                                                                                                                                                                                                                                                                                                           |                |
|------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------|
| <b>Q/V</b>                   | <b>Solution/Oplossing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | <b>Explanation/Verduideliking</b>                                                                                                                                                                                                                                                                                                         | <b>T&amp;L</b> |
| 4.1.1                        | $\text{Radius} = \frac{20}{2} \text{ mm} \quad \checkmark \text{MA}$ $= 10 \text{ mm} \quad \checkmark \text{A} \quad \textbf{OR/OF} \quad 1 \text{ cm}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | 1MA diameter by 2<br><br>1A simplification<br><b>AO</b><br><br>(2)                                                                                                                                                                                                                                                                        | M<br>L1<br>E   |
| 4.1.2                        | $\text{Height/Hoogte} = 309 \text{ mm} + 2(40 \text{ mm}) \quad \checkmark \text{C} \quad \checkmark \text{MA}$ $= 389 \text{ mm} \quad \checkmark \text{CA}$ $\approx 400 \text{ mm} \quad \checkmark \text{R}$                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1C convert to mm<br>1MA addition<br>1CA simplification<br><br>1R rounding<br><br>(4)                                                                                                                                                                                                                                                      | M<br>L2<br>M   |
| 4.1.3                        | <p>Total surface area ONE cork /Totale buite opp. EEN proppe</p> $= 2 \times 3,142 \times (10 \text{ mm})^2 + 2 \times 3,142 \times 10 \text{ mm} \times 43 \text{ mm} \quad \checkmark \text{SF}$ $= 628,4 \text{ mm}^2 + 2\,702,12 \text{ mm}^2$ $= 3\,330,52 \text{ mm}^2 \quad \checkmark \text{CA}$ <p>TSA for 3 layers on 53 corks /BO vir 3 lae op 53 proppe</p> $= 3\,330,52 \text{ mm}^2 \times 53 \times 3 \quad \checkmark \text{MCA}$ $= 529\,552,68 \text{ mm}^2$ <p>Convert to m<sup>2</sup>/Herlei na m<sup>2</sup> = <math>529\,552,68 \text{ mm}^2 \div 1\,000\,000</math> <math>\checkmark \text{C}</math></p> $= 0,52955... \text{ m}^2$ <p>Paint needed/Verf benodig : <math>\frac{0,52955 \text{ m}^2}{3 \text{ m}^2} \times 1 = 0,176... \ell \quad \checkmark \text{MCA}</math></p> $\approx 0,18 \ell$ <p><math>\therefore 0,18 \ell \times 1\,000 = 180 \text{ m}\ell \quad \checkmark \text{C}</math></p> <p>Total cost/Totale koste = <math>2 \times \text{R}55,00</math></p> $= \text{R}110,00 \quad \checkmark \text{CA}$ <p style="text-align: center;"><b>OR/OF</b></p> | <p><b>CA from 4.1.1</b></p> 1SF substitution in formula<br><br>1CA simplification<br><br>1MCA multiply by 53<br>1MCA multiply by 3<br><br>1C conversion to m <sup>2</sup><br><br>1MCA divide by 3 to get the number of litres<br><br>1C conversion to mℓ<br><br>1CA simplification<br><br><p style="text-align: center;"><b>OR/OF</b></p> | M/F<br>L3<br>D |

| Q/V | Solution/Oplossing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | T&L |
|-----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|     | <p>Total surface area ONE cork /Totale buite opp. EEN prop<br/>✓SF</p> $SA = 2 \times 3,142 \times (1 \text{ cm})^2 + 2 \times 3,142 \times 1 \text{ cm} \times 4,3 \text{ cm}$ $= 33,3052 \text{ cm}^2 \quad \checkmark \text{ CA}$ <p>SA of 53 corks/BO van 53 proppe = <math>33,3052 \text{ cm}^2 \times 53</math><br/>= 1 765,1756 ✓MC</p> <p>Convert to m/Herlei na m = <math>\frac{1\,765,1756}{10\,000}</math> ✓C<br/>= 0,17651756 m<sup>2</sup></p> <p>Paint needed(1 coat)/Verf benodig(1 laag)<br/>3 m<sup>2</sup> = 1 litre<br/>0,17651756 m<sup>2</sup> = x litres<br/><math>\frac{3x}{3} = \frac{0,17651756}{3}</math><br/>x = 0,0588391 litres ✓MCA</p> <p>Convert to mℓ/Herlei na mℓ = <math>0,0588391 \times 1\,000</math><br/>= 58,8391 ✓C</p> <p>Total amount of paint/Totale hoeveel. verf = <math>58,8391 \times 3</math><br/>= 176,51756 mℓ ✓MCA</p> <p>Number of cans of paint/Aantal blikke verf<br/>= <math>176,51756 \text{ mℓ} \div 125 \text{ mℓ}</math><br/>= 1,41214</p> <p>∴ Total cost/Totale koste = <math>2 \times \text{R}55</math><br/>= R110 ✓CA</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Total surface area ONE cork /Totale buite opp. EEN prop<br/>✓SF ✓C</p> $= [2 \times 3,142 \times (0,01 \text{ m})^2] + 2 \times 3,142 \times 0,01 \text{ m} \times 0,043 \text{ m}$ $= 0,00333052 \text{ m}^2 \quad \checkmark \text{ CA}$ <p>TSA 53 corks/TBO 53 proppe = <math>0,00333052 \text{ m}^2 \times 53 \times 3</math><br/>= 0,52955268 m<sup>2</sup> ✓CA ✓MCA</p> <p>Number of litres/Aantal liters = <math>\frac{0,52955268 \text{ m}^2}{3}</math> ✓MCA<br/>= 0,17651756 litres × 1 000<br/>= 176,51756 mℓ ✓C</p> <p>Number of cans/Aantal blikke = <math>\frac{176,51756 \text{ mℓ}}{125 \text{ mℓ}}</math><br/>= 1,412</p> <p>Cost/Koste = <math>2 \times \text{R}55</math><br/>= R110 ✓CA</p> <p style="text-align: center;"><b>OR/OF</b></p> | <p>1SF substitution in formula</p> <p>1CA simplification<br/>1MCA multiply by 53</p> <p>1C conversion to m<sup>2</sup></p> <p>1MCA divide by 3 to get the number of litres</p> <p>1C conversion to mℓ</p> <p>1MCA multiply 3</p> <p>1CA simplification</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1SF substitution in formula<br/>1C conversion to m<br/>1CA simplification</p> <p>1MCA multiply by 53 and 3</p> <p>1CA simplification</p> <p>1MCA divide by 3 to get the number of litres</p> <p>1C conversion to mℓ</p> <p>1CA simplification</p> <p style="text-align: center;"><b>OR/OF</b></p> |     |

| Q/V | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | T&L |
|-----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-----|
|     | <p>Total surface area ONE cork /Totale buite opp. EEN prop<br/> <math>\checkmark</math> SF<br/> <math>= 2 \times 3,142 \times (10 \text{ mm})^2 + 2 \times 3,142 \times 10 \text{ mm} \times 43 \text{ mm}</math><br/> <math>= 628,4 \text{ mm}^2 + 2\,702,12 \text{ mm}^2</math><br/> <math>= 3\,330,52 \text{ mm}^2 \checkmark</math> CA</p> <p><math>= 3\,330,52 \div (1\,000)^2 = 0,00333052 \text{ m}^2 \checkmark</math> C</p> <p>TSA for 3 layers/TBO vir 3 lae <math>= 0,00333052 \text{ m}^2 \times 3</math><br/> <math>= 0,009992 \text{ m}^2 \checkmark</math> MCA</p> <p>TSA(53 corks)/TBO(53 proppe) <math>= 0,009992 \text{ m}^2 \times 53</math><br/> <math>= 0,52955 \dots \text{ m}^2 \checkmark</math> MCA</p> <p>Paint/Verf: 1 ℓ covers/dek <math>3 \text{ m}^2</math></p> <p>1 tin/blik = 125 ml covers/dek <math>\frac{125 \text{ ml}}{1\,000 \text{ ml}} \times 3 \text{ m}^2 \checkmark</math> C<br/> <math>= 0,375 \text{ m}^2 \checkmark</math> MCA</p> <p>Number of cans/Aantal blikke <math>= \frac{0,52955 \text{ m}^2}{0,375 \text{ m}^2} = 1,41 \dots</math></p> <p>Total cost/Totale koste <math>= 2 \times \text{R}55,00</math><br/> <math>= \text{R}110,00 \checkmark</math> CA<br/> <b>OR/OF</b></p> <p>Radius <math>= 10 \text{ mm} = \frac{10}{1\,000} \text{ m} = 0,01 \text{ m}</math></p> <p>Height/Hoogte <math>= 43 \text{ mm} = \frac{43}{1\,000} \text{ m} = 0,043 \text{ m} \checkmark</math> C</p> <p>Total Surface Area/Buite-oppervlakte <math>\checkmark</math> SF<br/> <math>= 2 \times 3,142 \times (0,01)^2 + 2 \times 3,142 \times 0,01 \text{ m} \times 0,043 \text{ m}</math><br/> <math>= 0,00333052 \text{ m}^2 \checkmark</math> CA</p> <p>Spread rate/Spreikoers <math>3 \text{ m}^2 / \ell</math></p> <p>Number of litre/Aantal liters <math>= \frac{0,00333052 \text{ m}^2}{3 \text{ m}^2} \times 1 \checkmark</math> MCA<br/> <math>= 0,0011101733 \ell</math></p> <p>3 layers/3 lae <math>= 0,0011101733 \text{ litres} \times 3 \checkmark</math> MCA<br/> <math>= 0,0033305199 \text{ litres}</math></p> <p>For 53 corks/Vir 53 proppe <math>\checkmark</math> C<br/> <math>= 0,0033305199 \times 1000 \text{ ml} \times 53 \checkmark</math> MCA<br/> <math>= 176,5176 \text{ ml}</math></p> <p>Total cost/Totale koste <math>= 2 \times \text{R}55</math><br/> <math>= \text{R}110 \checkmark</math> CA</p> | <p>1SF substitution in formula</p> <p>1CA simplification</p> <p>1C conversion to <math>\text{m}^2</math></p> <p>1MCA multiply by 3</p> <p>1MCA multiply by 53</p> <p>1C conversion ml to ℓ<br/>1MCA coverage for <math>3 \text{ m}^2</math></p> <p>1CA simplification<br/><b>OR/OF</b></p> <p>1C conversion to m</p> <p>1SF substitution in formula</p> <p>1CA simplification</p> <p>1MCA divide by 3 to get the number of litres</p> <p>1MCA multiply by 3</p> <p>1C conversion to ml<br/>1MCA multiply by 53</p> <p>1CA simplification</p> | (8) |

| Q/V   | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                               | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | T&L           |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 4.2   | <p>Inside width of board/<i>Binnebreedte van bord</i><br/> <math>= 37,2 \text{ cm} - 2(4 \text{ cm})</math> ✓MA<br/> <math>= 29,2 \text{ cm}</math> ✓CA<br/> <math>= 292 \text{ mm}</math> ✓C</p> <p>Length of 4 patterns/<i>Lengte van 4 patrone</i><br/> <math>= (43 \text{ mm} + 20 \text{ mm}) \times 4</math> ✓MA<br/> <math>= 63 \text{ mm} \times 4</math><br/> <math>= 252 \text{ mm}</math> ✓CA</p> <p>INVALID/NIE GELDIG NIE ✓O</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Inside width of board/<i>Binnebreedte van bord</i><br/> <math>= 37,2 \text{ cm} - 2(4 \text{ cm})</math> ✓MA<br/> <math>= 29,2 \text{ cm}</math> ✓CA</p> <p>Length of pattern/<i>Lengte van patroon</i><br/> <math>= 43 \text{ mm} + 20 \text{ mm} = 63 \text{ mm}</math><br/> <math>= 6,3 \text{ cm}</math> ✓C</p> <p>Number of repeats needed to complete the row/<br/> <i>Getal patrone nodig om die ry te voltooi</i><br/> <math>= \frac{29,2 \text{ cm}}{6,3 \text{ cm}}</math> ✓MCA<br/> <math>= 4,63</math> ✓CA</p> <p>INVALID/NIE GELDIG NIE. ✓O</p> <p style="text-align: center;"><b>OR/OF</b><br/> ✓MA</p> <p>Board/<i>Bord</i>: <math>372 \text{ mm} - 80 \text{ mm} = 292 \text{ mm}</math> ✓CA<br/> Space needed for exactly 4 repeats/<i>Ruimte benodig om presies 4 keer te herhaal</i><br/> ✓MCA<br/> <math>= 292 \text{ mm} \div 4 = 73 \text{ mm}</math> ✓CA</p> <p>Each pattern/<i>Elke patroon</i> <math>= 43 \text{ mm} + 20 \text{ mm}</math><br/> <math>= 63 \text{ mm}</math> ✓A</p> <p>INVALID/NIE GELDIG NIE. ✓O</p> | <p>1MA subtraction<br/> 1CA simplification<br/> 1C conversion to mm</p> <p>1MA multiplying by pattern<br/> 1CA simplification</p> <p>1O opinion</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1MA subtraction<br/> 1CA simplification</p> <p>1C conversion to cm</p> <p>1MCA dividing by 6,3<br/> 1CA simplification<br/> 1O conclusion</p> <p style="text-align: center;"><b>OR/OF</b><br/> 1MA subtraction<br/> 1CA simplification</p> <p>1MCA dividing by 4<br/> 1CA simplification</p> <p>1A simplification<br/> 1O conclusion</p> <p style="text-align: right;">(6)</p> | MP<br>L4<br>M |
| 4.3.1 | <p>Number of lockers/<i>Aantal sluitkassies</i> <math>= 12 \times 5</math> ✓MA<br/> <math>= 60</math> ✓A</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Number of lockers/<i>Aantal sluitkassies</i> <math>= 6 \times 10</math> ✓MA<br/> <math>= 60</math> ✓A</p>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | <p>1MA multiplication<br/> 1A correct answer</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1MA multiplication<br/> 1A correct answer<br/> <b>AO</b></p> <p style="text-align: right;">(2)</p>                                                                                                                                                                                                                                                                                                                                                                                | MP<br>L1<br>E |

| Q/V           | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          | Explanation/Verduideliking                                                           | T&L           |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
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| 4.3.2         | To store extra books/lunch boxes/small sport bags/<br>extra clothes or shoes/<br>Keep learners' possessions safe. ✓✓O<br><i>Om ekstra boeke/kosblikke/klein sportsakke/ekstra<br/>klere of skoene te stoor/<br/>Hou besittings van leerders veilig.</i>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     | 2O reason<br><br>(ACCEPT ANY ONE)<br>(2)                                             | MP<br>L4<br>E |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| 4.3.3<br>(a)  | Volume = L × w × h/Volume = L × b × h ✓SF<br>32 400 cm <sup>2</sup> = 45 cm × B × 18 cm<br><br>B = $\frac{32\ 400\text{ cm}^3}{45\text{ cm} \times 18\text{ cm}}$ ✓MA<br>= 40 cm ✓CA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        | 1SF substitution<br><br>1MA changing subj of<br>formula<br>1CA simplification<br>(3) | M<br>L2<br>M  |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| 4.3.3<br>(b)  | Total length of one stack/Totale lengte van een stapel<br>✓MCA ✓C<br>(2 × 40 cm) + 5,7 cm + (2,8 cm × 2)<br>= 91,3 cm ✓CA<br><br>Total length of a set/Totale lengte van 'n stel<br>✓MCA ✓MA<br>(91,3 cm × 5) + (4 × 0,2 cm)<br>= 456,5 cm + 0,8 cm<br>= 457,3 cm ✓CA<br><br><b>OR/OF</b><br><br>Length of one stack/ Lengte van een stapel<br>✓C ✓MCA<br>= 2(400 mm) + 2(28 mm) + 57 mm<br>= 913 mm ✓CA<br><br>Total length of a set/Totale lengte van 'n stel<br>= (913 mm × 5) + (4 × 2 mm) ✓MCA ✓MA<br>= 4 565 mm + 8 mm<br>= 4 573 mm ✓CA<br><br><b>OR/OF</b><br><br>Total lengths/Totale lengtes<br><table><tr><td></td><td>In cm</td><td>In mm</td></tr><tr><td>A:</td><td>2,8 × 10 ✓C</td><td>28 × 10 ✓MCA</td></tr><tr><td>B:</td><td>40 × 10 ✓MCA</td><td>400 × 10 ✓C</td></tr><tr><td>C:</td><td>5,7 × 5 ✓MCA</td><td>57 × 5 ✓MCA</td></tr><tr><td>Space/Spasie:</td><td>0,2 × 4 ✓MA</td><td>2 × 4 ✓MA</td></tr></table><br>✓MCA<br>Total/Totaal = 400 cm + 28 cm + 28,5 cm + 0,8 cm<br>= 457,3 cm ✓CA<br><b>OR/OF</b> ✓MCA<br>Total/Totaal = 4000 mm + 280 mm + 285 mm + 8 mm<br>= 4 573 mm ✓CA |                                                                                      | In cm         | In mm | A: | 2,8 × 10 ✓C | 28 × 10 ✓MCA | B: | 40 × 10 ✓MCA | 400 × 10 ✓C | C: | 5,7 × 5 ✓MCA | 57 × 5 ✓MCA | Space/Spasie: | 0,2 × 4 ✓MA | 2 × 4 ✓MA | <b>CA from 4.3.3 (a)</b><br><br>1MCA multiply by 2<br>1C conversion to cm<br>1CA length in cm<br><br>1MCA multiplication<br>1MA adding in between<br>spaces<br>1CA simplification<br><b>OR/OF</b><br><br>1C conversion to mm<br>1MCA multiply by 2<br>1CA length in cm<br><br>1MCA multiplication<br>1MA adding in between<br>spaces<br>1CA simplification<br><b>OR/OF</b><br><br>1C conversion to cm<br>1MCA multiply by 10<br>1MCA multiply by 5<br>1MA spaces in between<br><br>1MCA adding<br>1CA simplification<br><b>OR/OF</b><br>1MCA adding<br>1CA simplification<br>(6) | M<br>L2<br>D |
|               | In cm                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | In mm                                                                                |               |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| A:            | 2,8 × 10 ✓C                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 28 × 10 ✓MCA                                                                         |               |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| B:            | 40 × 10 ✓MCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 400 × 10 ✓C                                                                          |               |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| C:            | 5,7 × 5 ✓MCA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                | 57 × 5 ✓MCA                                                                          |               |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
| Space/Spasie: | 0,2 × 4 ✓MA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2 × 4 ✓MA                                                                            |               |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |
|               |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                             | [33]                                                                                 |               |       |    |             |              |    |              |             |    |              |             |               |             |           |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |              |

| <b>QUESTION 5 [28 MARKS]</b> |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |                                                                                                                                                                                                                                                                                                                                                                                                                                                                            |                |
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| <b>Q/V</b>                   | <b>Solution/Oplossing</b>                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | <b>Explanation/Verduideliking</b>                                                                                                                                                                                                                                                                                                                                                                                                                                          | <b>T&amp;L</b> |
| 5.1.1                        | 2 Feb ✓✓RT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           | 2RT correct date<br>(2)                                                                                                                                                                                                                                                                                                                                                                                                                                                    | M<br>L1<br>M   |
| 5.1.2                        | Highest wave – Lowest wave/ <i>Hoogste golf – laagste golf</i><br>✓RT ✓RT<br>= 2,27 m – 0,09 m<br>= 2,18 m ✓CA                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                       | 1RT 2,27<br>1RT 0,09<br>1CA simplification<br>(3)                                                                                                                                                                                                                                                                                                                                                                                                                          | M<br>L2<br>M   |
| 5.1.3                        | Time duration/ <i>Tydsduur</i> : 06:46 to 17:25 ✓RT<br><br>06:46 to/tot 07:00 14 minutes/ <i>minute</i><br>07:00 to/tot 12:00 5 hours/ <i>uur</i> ✓MA<br>12:00 to/tot 17:25 5 hours/ <i>uur</i> 25 minutes/ <i>minute</i><br>✓CA<br>Total time = 10 hours and 39 minutes<br><i>Totale tydsduur = 10 uur 39 min</i><br>INCORRECT/ <i>NIE KORREK NIE.</i> ✓O<br><br><b>OR/OF</b><br><br>Sunset time/ <i>Sonsondergangstyd</i><br>✓RT<br>= 6:46 + 10 hours/ <i>uur</i> 48 min<br>= 16:46 + 48 min ✓MA<br>= 17:34 ✓CA<br><br>INCORRECT/ <i>NIE KORREK NIE.</i> ✓O<br><b>OR/OF</b><br>From/ <i>Vanaf</i> 06:46 to/tot 17:25<br>Duration/ <i>Tydsduur</i> = 17h25 – 06h46 ✓R<br>= (17 × 60 min + 46 min) – (6 × 60 min + 25)<br>= 1 045 min – 406 min<br>= $\frac{639 \text{ min}}{60}$ ✓MA<br>= 10 hours and 39 min ✓CA<br>INCORRECT/ <i>NIE KORREK NIE</i> ✓O<br><br><b>OR/OF</b><br>6:46 am to 5:25 pm = 17:25 ✓RT<br><br>∴ 17:25 – 6:46 ✓MA ✓CA<br>= 10 hours 39 minutes / <i>10 ure 39 min</i><br>INCORRECT/ <i>NIE KORREK NIE</i> ✓O | 1RT reading correct values<br><br>1MA finding duration<br>1CA simplification<br><br>1O conclusion<br><br><b>OR/OF</b><br><br>1RT reading correct values<br><br>1MA adding duration<br>1CA simplification<br><br>1O comparison<br><b>OR/OF</b><br><br>1RT reading correct values<br><br>1MA finding duration<br>1CA simplification<br>1O conclusion<br><br><b>OR/OF</b><br>1RT reading correct values<br>1MA finding duration<br>1CA simplification<br>1O conclusion<br>(4) | M<br>L4<br>M   |

| Q/V          | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                       | T&L           |
|--------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|---------------|
| 5.2.1        | 4 ✓✓A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 | 2A correct number of airports<br>(2)                                                                                                                                                                                                                                                                                                                                             | MP<br>L2<br>E |
| 5.2.2<br>(a) | <p>Length of the bar scale/<i>Lengte van staafskaal</i>:<br/> <math>= 20 \text{ mile/myl} \div 0,621371 \quad \checkmark C</math><br/> <math>= 32,1868\dots</math><br/> <math>\approx 32,2 \text{ km} \quad \checkmark CA</math></p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Scale in km/<i>Skaal in km</i><br/> <math>3,5 \text{ cm} = 20 \text{ miles/myl} \div 0,621371 \quad \checkmark C</math><br/> <math>3,5 \text{ cm} = 32,18688\dots \quad \checkmark CA</math><br/> <math>1 \text{ cm} : 9,1962\dots \text{km}</math><br/> <math>\approx 1 \text{ cm} : 9 \text{ km}</math></p> <p style="text-align: center;"><b>OR/OF</b></p> <p><math>1 \text{ km} = 0,621371 \text{ miles/myl}</math><br/> <math>1 \div 0,621371 = 1,6093444</math></p> <p>Length of the bar scale/<i>Lengte van staafskaal</i>:<br/> <math>1,609344498 \times 20 \quad \checkmark C</math><br/> <math>= 32,186888</math><br/> <math>\approx 32,2 \text{ km} \quad \checkmark CA</math></p> | <p>1C convert to km<br/> 1CA simplification</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1C convert to km<br/> 1CA simplification</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1C convert to km<br/> 1CA simplification<br/> (2)</p>                                                                                                                         | M<br>L2<br>E  |
| 5.2.2<br>(b) | <p>✓A<br/> 35 mm is 32,2 km<br/> 65 mm is <math>n</math> km<br/> <math>n = \frac{65}{35} \times 32,1868\dots \quad \checkmark MCA</math><br/> <math>= 59,7756\dots \text{ km}</math><br/> <math>\approx 59,8 \text{ km} \quad \checkmark CA</math></p> <p>VALID/<i>GELDIG</i> ✓O</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>If 5.2.2(a) scale is <math>1 \text{ cm} : 9,196254\dots \quad \checkmark CA</math><br/> <i>Indien 5.2.2(a) skaal <math>1 \text{ cm} : 9,196254\dots</math> is</i></p> <p>Distance/<i>Afstand</i> <math>= 6,5 \text{ cm} \times 9,196254\dots \quad \checkmark MCA</math><br/> <math>= 59,77\dots \quad \checkmark CA</math></p> <p>VALID/<i>GELDIG</i> ✓O</p>                                                                                                                                                                                                                                                                                 | <p><b>CA from 5.2.2(a)</b><br/> 1A measured distance</p> <p>1MCA using the scale<br/> 1CA simplification<br/> 1O conclusion</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1CA measured distance from 5.2.2(a)</p> <p>1MCA using the scale<br/> 1CA simplification<br/> 1O conclusion<br/> <b>[Accept 33 mm – 36 mm]</b><br/> <b>[Accept 58 km – 64 km]</b><br/> (4)</p> | MP<br>L4<br>M |

| Q/V   | Solution/Oplissing                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   | Explanation/Verduideliking                                                                                                                                                                                                                                                                                                                                                                                              | T&L                     |
|-------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|-------------------------|
| 5.3.1 | <p>Time /Tyd :<br/> 08:50 to/tot 9:21 ✓RT<br/> = 31 min ✓A<br/> Distance = Speed × Time/Afstand = <i>Spoed</i> × <i>Tyd</i><br/> 21,3 miles/<i>myl</i> = Speed/<i>Spoed</i> × 31 min ✓SF ✓C<br/> 21,3 miles/<i>myl</i> = Speed/<i>Spoed</i> × 0,5166666... hours</p> <p>Speed in miles/hour/<i>Spoed in myl/uur</i></p> $= \frac{21,3}{0,516666667}$ <p>= 41,22581 ✓CA</p>                                                                                                                                                                                                                                                                           | <p>1RT correct times<br/> 1A duration<br/> 1SF substitution<br/> 1C conversion to hours</p> <p>1CA simplification<br/> (5)</p>                                                                                                                                                                                                                                                                                          | <p>M<br/> L3<br/> M</p> |
| 5.3.2 | <p>✓RT<br/> 25 miles/<i>myl</i> = 1 gallon/<i>gelling</i> ✓MA<br/> 21,3 miles × 0,04 gallon per mile = 0,852 gallons</p> <p>Total cost of fuel/<i>Brandstofkoste</i><br/> ✓RT<br/> = 0,852 × \$4,757 ✓MA<br/> = \$4,05 ✓CA</p> <p>CORRECT/KORREK. ✓O</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>Fuel cost/<i>Brandstofkoste</i> \$4,757 per gallon /<i>gelling</i><br/> SUV uses/<i>verbruik</i> 25 mpg</p> <p>Cost for/<i>Koste vir</i> 25 mile/<i>myl</i> = \$4,757<br/> ✓RT ✓RT<br/> ✓MA ∴ 21,3 miles/<i>myl</i> = <i>n</i><br/> <math>n = \frac{21,3}{25} \times \\$4,757</math> ✓MA<br/> ≈ \$4,05 ✓CA</p> <p>CORRECT/KORREK. ✓O</p> | <p>1RT correct value 25 mpg<br/> 1MA number of gallons</p> <p>1RT price per gallon<br/> 1MA multiply gallons with cost<br/> 1CA simplification</p> <p>1O conclusion</p> <p style="text-align: center;"><b>OR/OF</b></p> <p>1RT price per gallon<br/> 1RT correct value 25 mpg<br/> 1MA number of gallons <math>\frac{21,3}{25}</math><br/> 1MA multiplication<br/> 1CA simplification</p> <p>1O conclusion<br/> (6)</p> | <p>M<br/> L4<br/> D</p> |
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | [28]                                                                                                                                                                                                                                                                                                                                                                                                                    |                         |
|       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      | <b>TOTAL: 150</b>                                                                                                                                                                                                                                                                                                                                                                                                       |                         |